

PROJECT PROPOSAL

Intelligent Reasoning Systems Project

- **Project Title:** REITerrätsel - Equity Risk Solver for S-REITs
- **Group Number:** Group 16 (Canvas)
- **Group Members:** Jason Tay Neng Wei (A0265092A)
- **Date of Presentation:** Unconfirmed
- **Date of Submission:** 27 March 2026
- **Sponsor/Client:** NA

Introduction

I aim to design and implement a system that predicts vulnerability of dividend investment vehicles (specifically Singapore Real-Estate Investment Trusts (S-REITs)), based on macroeconomic factors and company financials. The project aims to provide actionable feedback for Singaporean investors to improve decision-making when it comes to prioritizing tickers.

Project Background / Market Context

There are a variety of public, simple yield screens when selecting S-REITs available to Singapore retail investors.

However, there is a lack of quantitative evaluation available that captures the safety of current S-REITs, or whether current distributions remain sustainable under changing policy and inflation. This project addresses that gap by building intelligent reasoning systems that combine macro regime signals with REIT-level financial health indicators.

The main challenge is to map macroeconomic shocks and accounting values into practical investment risk signals. Aside from commonly computed metrics such as dividend cut vulnerability, gearing pressure, and interest coverage stress, this project aims to use Cognitive Techniques, Knowledge Discovery, and Knowledge-Based Reasoning to refine rulesets and eliminate ineffective thresholds.

The practical market context has some significant concerns:

- **Recurrent risks:** Debt cost sensitivity, Refinancing risk, Valuation compression
- **Unseen Risks:** Regulatory floor revisions like the Code on Collective Investment Schemes have decoupled leverage limits from interest coverage requirements.

Metric	Previous Requirement (for >45% Leverage)	New Requirement (for All REITs)
Min. Interest Coverage Ratio (ICR)	2.5 times	1.5 times
Maximum Aggregate Leverage	50%	50%

BDO Spotlight: REIT Leverage and Disclosure [2]

This means a S-REIT operating near the regulatory floor today offers materially thinner protection against rate shocks than before.

The project therefore prioritizes mechanisms that are operationally interpretable, and that can surface where individual S-REITs sit relative to these consequential events.

Literature Review / Market Research

Academic Research (Research-Oriented)

Key method reference: Ratner et al. (2017), *Snorkel: Rapid Training Data Creation with Weak Supervision* [1].

Snorkel enables users to train models without hand-labelling any training data. Instead, labelling functions expressing heuristics with unknown accuracies and correlations are combined via a generative model. This is directly applicable here because ground-truth distress labels do not exist for S-REITs. This serves as an alternative and falsification method to hand-labelling rules.

The generative model learns the accuracies and correlations of labelling functions without ground truth labels. It combines them to approximate the ground truth using probabilities. This means Snorkel's conflict matrix (not its accuracy estimates, due to the limited size of the dataset/time periods) is the primary output of value.

Market Research (MVP Product-Oriented)

1. **REITsavvy Screener** offers screening across 19 criteria including gearing ratio, P/NAV, occupancy rate, and WALE, with proprietary ratings and fundamental analysis reports for 38 S-REITs. REITsavvy exposes raw metrics and lets users filter manually, without any form of knowledge engineering applied.
2. **Fifth Person** provides live daily S-REIT data including gearing ratios (inclusive of perpetual securities) for general informational purposes. It focuses on pure data display with no rule-based scoring, MAS thresholds, or distress ranking output. Retail investors still bear the burden of interpreting the numbers themselves.
3. **DBS Research InsightsDirect** hosts an analyst-produced, episodic system that requires institutional access and cannot be re-run by a retail investor on current data.

Existing tools either display metrics without reasoning (REITsavvy, Fifth Person) or apply reasoning episodically behind institutional paywalls (DBS Research). This project encodes MAS regulatory rules as sequenced, machine-validated logic and exposes distress rankings programmatically, which alternatives do not.

Project Scope

Knowledge-Based Reasoning (Decision Automation)

- As a prior, MAS regulatory thresholds are hand-encoded as a structured production rule system
- Snorkel resolves conflicts between rules using a generative model, acting as an automated reasoner over the KB
- Rules obtained can be used to query for the current state of a S-REIT (healthy, distressed, etc.)

Knowledge Discovery (Data Mining)

- Snorkel applies the expert KB to unlabelled REIT-year rows to generate probabilistic training labels.
- The Decision Tree trains on these labels to extract which variables the data surfaces as primary risk discriminators
- This extracts/mines additional structured labels from domain heuristics without hand-labelling (the rules are induced from the data)

Cognitive Techniques / KG

- Final rules and S-REIT evaluation results are loaded into Neo4j as a knowledge graph with nodes (REITs, Rules, Labels) and edges (TRIGGERS, PRODUCES)
- Streamlit queries the graph using Cypher
- The UI exposes distress rankings in plain query form

Process Build a S-REIT risk evaluation system for a curated set of tickers only.

- Combine company financials with macroeconomic indicators (See Appendices A and B).

- Operationalise hard risk flags as rule-based systems (MAS regulatory thresholds such as gearing, ICR, FFO payout).
- Focus on practical decision support for a retail investor context.

Reasoning focus:

- Hybrid architecture:
 - Hand-crafted Rule Knowledge Base (KB) (MAS / financial thresholds)
 - Machine-induced rule extraction (Snorkel probabilistic labelling -> Decision Tree)
 - Combines the previous two components to evaluate validity of rules.
- Use the KB resulting from the aforementioned architecture to rank S-REIT distress severity.
- Rules are sequenced and conflict-resolved. (Use directional/trend rules to supplement level-based rules)
- Knowledge graph (Neo4j) encodes final rules and evaluation results for programmatic inference
- Streamlit UI exposes distress rankings to end users.

Out of scope / constraints:

- Data limitations. S-REITs specifically are a young asset class. The SGX REIT market only matured post-2010, and several tickers listed even later.
- Price prediction will not be part of this project.
- Time Series/LSTM modeling will not be part of this project.

Data Collection and Preparation

Dataset Overview

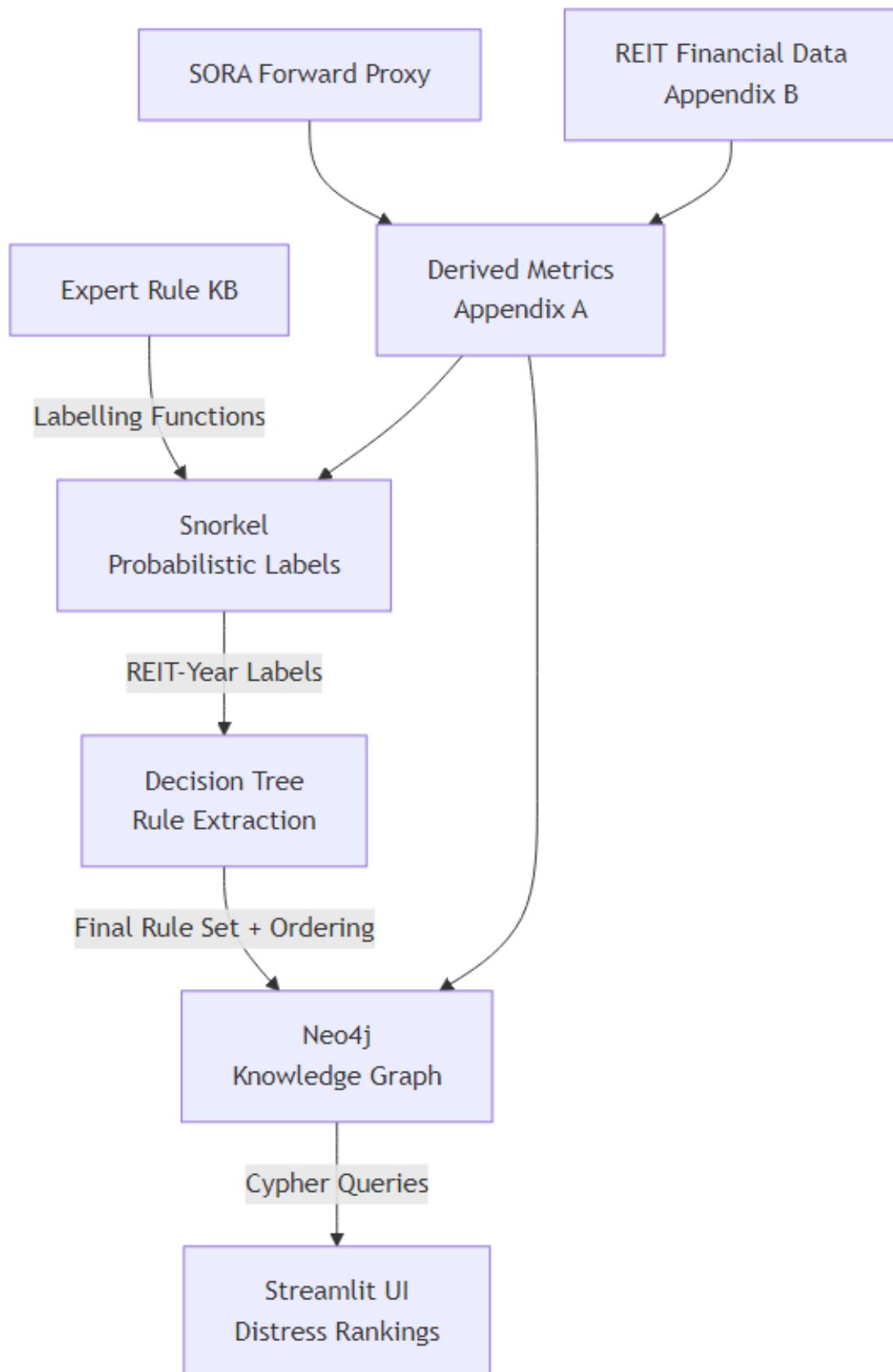
I will prefer free or open-source data for this project:

- Scraped Exchange Financial Statements (S-REITs)
- Forward-facing SG Rate (SORA) proxies:
 - 1-Year Expected Inflation (EXPINF1YR)
 - MAS SGS short-term T-bills yield (6-month and 1-year)
 - Fed Funds Decentralized Forecasts
- Historical data:
 - MAS Compounded 3M SG Rate (SORA) Series
 - Historical PI references (US BLS / Singapore DOS)
 - Historical SGS 10Y movement data (long-tail)

Note:

- Normalization is required for financial data obtained from exchange financial statements as it alternates between annual and quarterly data.
- 1, 2 are mandatory data for the implementation of this project. 3 is optional data.
- Further elimination of data will be made to determine the most effective Datasets.

System Design (if available)



System Architecture Flow

1. Expert rule KB is encoded as labelling functions.
2. Snorkel generates probabilistic REIT-year labels.
3. Decision Tree extracts final rule ordering.
4. Final rules and evaluations are loaded into Neo4j.
5. Streamlit queries Neo4j to present distress rankings.

Platform Layers

- **Application Layer:** Streamlit
- **Data Layer:** CSV / DuckDB

Data Core Computations

Rank	Group	Components
1	Debt Structure & Refinancing Risk	Refinancing Risk Ratio, Net Debt / EBITDA, Implicit Cost of Debt
2	Balance-Sheet Resilience	Gearing Ratio, ICR, Net Debt / EBITDA
3	Distribution Sustainability	FFO Coverage Margin, Capex Drag Diagnostic, FFO YoY Growth
4	Income Quality & Trend	FFO YoY Growth, Non-Recurring Income Share, OpEx Intensity
5	Equity Stress Signals	P/NAV Proxy (Price to Book), Unit Dilution Rate, Revenue Concentration
6	Return Quality	FFO Yield [Equity], FFO Yield [Capital], Leverage Premium
7	Real Yield	Real Yield (Opportunity Cost), Real Yield (Inflation)

Note: See the dataset overview in Appendices A and B.

Model Layer

- **Reasoning objective:** Combine expert priors with weak supervision to produce explainable distress logic under limited labelled data.
- **Validation objective:** Use decision-tree rule extraction to check whether data-supported split logic aligns with, refines, or challenges expert assumptions.
- **Explainability objective:** Persist rule lineage and triggered outcomes in Neo4j so Streamlit can return auditable distress rankings.
- **Optional extension:** XGBoost/Random Forest for targeted correlation checks where additional signal testing is needed.

Technique Selection Justification

Snorkel was selected as the weak-supervision layer in the absence of a large curated ground-truth dataset. The project starts with sparse labels and domain rules; compared with directly training supervised models or using simple rule voting, Snorkel provides a principled way to combine overlapping and conflicting expert labelling functions, which is better aligned with this small-N, expert-driven setting.

Decision Tree was selected for rule extraction because this stage prioritizes transparent, inspectable split logic over marginal predictive gains. `Sklearn` APIs were chosen because Orange does not allow for customization of tree-depth.

Neo4j was selected for final rule storage and query because the output is naturally graph-structured (REITs, metrics, triggered rules, and severity relationships) and requires relationship-centric traversal. Compared with a purely relational schema, Neo4j with Cypher reduces join-heavy query logic for explainability views (e.g., “which rule path caused this ranking”), while keeping rule lineage easier to audit.

Implementation (if available)

- Data will be scraped and aggregated based on the dimensions in Appendix A (yearly/quarterly, so normalization is needed).
- Rules are coded as Python labelling functions
- Snorkel outputs one probabilistic label per REIT-year row; piped into Decision Trees as training data
- Decision tree extracts final rule set with explicit split ordering (first split = most important discriminator)
- Evaluate outputted rules for validity (repeatable)
- Final Rules + S-REIT evaluation results written into Neo4j; Streamlit reads from Neo4j at query time

Results and Progress (if available)

- Data sources identified but not yet fully scraped
- Data pipeline and model layer in progress
- Snorkel-Rule patching system designed (pending implementation)

Challenges and Roadblocks (if available)

- Limited direct data for some ideal formulas (requires proxies)
- Mixed data frequencies across sources
- Mixed data source availability (shareholder releases (.pptx) vs exchange data (.csv))
 - Some S-REITs use non-calendar accounting years; screener data used as pragmatic substitute due to .pptx conversion limitations
- Small dataset (annual data means a maximum of N~84 usable rows) means Snorkel accuracy estimates serve as conflict-surfacing tools rather than robust accuracy estimators (statistically weak)

Future Work (if available)

- Implement additional S-REIT-specific data coverage, subject to **PPTX scraping limitations**.
 - Include:
 - * **Forward-Looking Interest Costs**
 - Include debt maturity details (e.g., 12/24-month profile), weighted average debt maturity (WADM), and fixed vs floating debt mix.
 - * **Cost Pass-Through / Lease Strength**
 - Include WALE (Weighted Average Lease Expiry), noting two weighting methods: by occupied area; by gross rental income (GRI), which is common and usually more conservative in S-REIT reporting.
 - Important caveat: WALE values are **not directly comparable** across S-REITs unless weighting methodology is normalized.
 - * **Operating and Tenant Risk Indicators**
 - Add rental reversion rates, occupancy rate, and tenant industry concentration.
 - * **Data Quality / Risk Note**
 - Tenant concentration may rely on Schema 05 revenue breakdown, which could be incomplete.
 - A **tail-risk flag** not captured elsewhere in the model.
- Broaden scope to more dividend equity types or more S-REIT tickers.
- Build an MLOps pipeline.
- Add more experts to the hybrid architecture.
- Apply common macroeconomic factors to test other investing strategies (Bogleheads, Momentum, Factor, etc.).

Conclusion (if available)

I aim to deliver an explainable S-REIT distress-ranking system that combines macro signals, firm-level financial metrics, and rule-based evaluation into a single decision workflow.

It directly fulfills the IRS technique groups defined in scope: **Knowledge-Based Reasoning** (MAS and financial-health production rules with conflict handling), **Knowledge Discovery** (Snorkel weak supervision and Decision Tree rule induction), and **Cognitive Techniques / KG** (Neo4j graph representation with Cypher-queryable rule lineage and outcomes).

Academically, the project contributes a reproducible hybrid reasoning pattern for low-label financial domains, where expert priors can be systematically compared, stress-tested, and refined by data-induced rules.

In market terms, it addresses a retail-investor pain point by converting fragmented screening data into transparent, repeatable distress signals, reducing reliance on ad-hoc interpretation and improving ability to prioritize S-REIT tickers under changing regimes.

Supplementary Materials

References

- [1] Ratner, A., Bach, S., Ehrenberg, H., Fries, J., Wu, S., and Re, C. (2017). *Snorkel: Rapid Training Data Creation with Weak Supervision*. Proceedings of the VLDB Endowment (PVLDB).
- [2] BDO Singapore (2025). *REIT Leverage and Disclosure*.

Appendix A: Derived Financial Data Index

1) Overview

Dataset	Purpose	Rows
Data Groupings	Group-level modeling buckets	7
Data Derived (Raw)	Metric-level definitions	20

2) Groupings

Rank	Group	Components
1	Debt Structure & Refinancing Risk	Refinancing Risk Ratio, Net Debt / EBITDA, Implicit Cost of Debt
2	Balance-Sheet Resilience	Gearing Ratio, ICR, Net Debt / EBITDA
3	Distribution Sustainability	FFO Coverage Margin, Capex Drag Diagnostic, FFO YoY Growth
4	Income Quality & Trend	FFO YoY Growth, Non-Recurring Income Share, OpEx Intensity
5	Equity Stress Signals	P/NAV Proxy (Price to Book), Unit Dilution Rate, Revenue Concentration
6	Return Quality	FFO Yield [Equity], FFO Yield [Capital], Leverage Premium
7	Real Yield	Real Yield (Opportunity Cost), Real Yield (Inflation)

3) Raw Metrics

Metric	Formula (short)	Schema
Interest Coverage Ratio (ICR)	EBITDA / Total Interest Expense	SCHEMA_01
Gearing Ratio	Debt to Assets Ratio	SCHEMA_02
Net Debt / EBITDA	Net Debt / EBITDA	SCHEMA_01
Refinancing Risk Ratio	Short Term Debt / Total Debt	SCHEMA_01
FFO YoY Growth	(FFO_t - FFO_t-1) / FFO_t-1	SCHEMA_01
Implicit Cost of Debt	Interest Paid / Total Debt	SCHEMA_01
Revenue Concentration (Top Segment Share)	(Largest segment revenue / Total revenue) or HHI: sum of (segment_i / total_rev)^2	SCHEMA_05
Payout Ratio	Total Cash Dividends Paid / FFO	SCHEMA_01
FFO Coverage Margin	(FFO - Total Cash Dividends Paid) / FFO	SCHEMA_01
P/NAV Proxy (Price to Book)	Price to Book Ratio	SCHEMA_02
FFO Yield [Equity]	FFO / (Net Income * PE Ratio)	SCHEMA_01, SCHEMA_02
FFO Yield [Capital]	FFO / Enterprise Value	SCHEMA_02
Leverage Premium	FFO Yield [Equity] - FFO Yield [Capital]	
Real Yield (Opportunity Cost)	Dividend Yield (FY) % - SORA Compounded 3M	SCHEMA_03
Real Yield (Inflation)	Dividend Yield (FY) % - Expected CPI	SCHEMA_03
Unit Dilution Rate	(Diluted Shares_t - Diluted Shares_t-1) / Diluted Shares_t-1	SCHEMA_01
OpEx Intensity	Total Operating Expenses / Total Revenue	SCHEMA_01
Capex Drag Diagnostic	(Total Cash Dividends Paid / FCF) > (Total Cash Dividends Paid / FFO)	SCHEMA_01
Non-Recurring Income Share	Unusual Income/Expense / Net Income	SCHEMA_01
DSCR	FFO / (Interest Paid + Short Term Debt)	SCHEMA_01

Appendix B: Core Data Sources (Schema Index)

1) Overview

Schema	Time Basis	Sections	Key Fields (sample)
SCHEMA_01_STATEMENTS	Annual	IS / BS / CF	Total revenue, Net income, Total assets, Total debt, Free cash flow
SCHEMA_02_STATISTICS	Quarterly	Valuation / Profitability / Liquidity / Solvency / Per-share	P/E, ROE %, Debt-to-equity, Revenue/share
SCHEMA_03_DIVIDENDS	Annual	Dividends	DPS (FY), Yield %, Payout ratio %
SCHEMA_04_EARNINGS	Quarterly	EPS / Revenue	Reported, Estimate, Surprise
SCHEMA_05_REVENUE	Annual	By Source / By Country	Revenue by asset, Revenue by region

Legend: IS = Income Statement, BS = Balance Sheet, CF = Cash Flow

2) Schema Details

SCHEMA_01_STATEMENTS (Annual)

Section	Level	Metric
Income Statement	L0	Total revenue // YoY growth
Income Statement	L1	Rental income
Income Statement	L1	Investment income
Income Statement	L1	Interest income (revenue)
Income Statement	L1	Other operating income
Income Statement	L1	Trading account income
Income Statement	L1	Trust & fiduciary income, commissions & fees
Income Statement	L2	Trust income
Income Statement	L2	Commision & fee income
Income Statement	L0	Total operating expenses
Income Statement	L1	Loan loss provision
Income Statement	L1	Selling/general/admin expenses (total)
Income Statement	L1	Cost of goods sold
Income Statement	L2	Depreciation and amortization
Income Statement	L2	Other cost of goods sold
Income Statement	L1	Other operating expenses (total)
Income Statement	L0	Operating income
Income Statement	L0	Non-operating income (excl. interest expenses)
Income Statement	L1	Non-operating interest income
Income Statement	L1	Miscellaneous non-operating expense
Income Statement	L1	Pretax equity in earnings
Income Statement	L0	Total interest expense (banks)
Income Statement	L0	Unusual income/expense
Income Statement	L0	Pretax income
Income Statement	L0	Taxes
Income Statement	L0	Equity in earnings
Income Statement	L0	After tax other income/expense
Income Statement	L0	Non-controlling/minority interest
Income Statement	L0	Net income before discontinued operations
Income Statement	L0	Preferred dividends
Income Statement	L0	Net income // YoY growth
Income Statement	L0	Basic earnings per share (basic EPS)
Income Statement	L0	Diluted earnings per share (diluted EPS) // YoY growth
Income Statement	L0	Average basic shares outstanding
Income Statement	L0	Diluted shares outstanding
Income Statement	L0	EBITDA // YoY growth
Income Statement	L0	EBIT // YoY growth
Balance Sheet	L0	Total assets // YoY growth
Balance Sheet	L1	Other long term assets (total)
Balance Sheet	L1	Intangible assets
Balance Sheet	L1	Deferred tax assets
Balance Sheet	L1	Net property/plant/equipment
Balance Sheet	L1	Investments in unconsolidated subsidiaries
Balance Sheet	L1	Total investments (banks)
Balance Sheet	L2	Net loans
Balance Sheet	L3	Gross Loans
Balance Sheet	L3	Loan loss allowances (reserves)
Balance Sheet	L2	Real estate assets
Balance Sheet	L2	Other investments
Balance Sheet	L1	Total receivables (net)
Balance Sheet	L1	Cash and short term investments
Balance Sheet	L2	Cash & equivalents

Section	Level	Metric
Balance Sheet	L2	Short term investments
Balance Sheet	L0	Total liabilities // YoY growth
Balance Sheet	L1	Other non-current liabilities (total)
Balance Sheet	L1	Deferred income (non-current)
Balance Sheet	L1	Deferred tax liabilities
Balance Sheet	L1	Provision for risks & charge
Balance Sheet	L1	Total debt
Balance Sheet	L2	Short term debt
Balance Sheet	L2	Long term debt
Balance Sheet	L3	Long term debt (excl. lease liabilities)
Balance Sheet	L3	Capital and operating lease obligations
Balance Sheet	L1	Total deposits
Balance Sheet	L0	Total equity // YoY growth
Balance Sheet	L1	Shareholders' equity
Balance Sheet	L2	Common equity (total)
Balance Sheet	L3	Retained earnings
Balance Sheet	L3	Additional paid-in capital / Capital su...
Balance Sheet	L4	Common stock par / Carrying value
Balance Sheet	L4	Treasury stock (common)
Balance Sheet	L3	Other common equity
Balance Sheet	L2	Preferred stock, carrying value
Balance Sheet	L1	Minority interest
Balance Sheet	L0	Total liabilities & shareholders' equities
Balance Sheet	L0	Net debt
Cash Flow	L0	Cash from operating activities // YoY growth
Cash Flow	L1	Funds from operations
Cash Flow	L1	Changes in working capital
Cash Flow	L2	Change in accrued expenses
Cash Flow	L2	Change in accounts receivable
Cash Flow	L2	Change in inventories
Cash Flow	L2	Change in accounts payable
Cash Flow	L2	Change in taxes payable
Cash Flow	L2	Change in other assets/liabilities
Cash Flow	L0	Cash from investing activities // YoY growth
Cash Flow	L1	Purchase/sale of business (net)
Cash Flow	L2	Purchase/acquisition of business
Cash Flow	L2	Sale of fixed assets & businesses
Cash Flow	L1	Purchase/sale of investments (net)
Cash Flow	L2	Purchase of investments
Cash Flow	L2	Sale/maturity of investments
Cash Flow	L1	Capital expenditures
Cash Flow	L2	Capital expenditures (fixed assets)
Cash Flow	L2	Capital expenditures (other assets)
Cash Flow	L1	Proceeds from loans
Cash Flow	L2	Increase in loans
Cash Flow	L2	Decrease in loans
Cash Flow	L1	Other investing cash flow items (total)
Cash Flow	L2	Investing activities (other uses)
Cash Flow	L2	Investing activities (other sources)
Cash Flow	L0	Cash from financing activities // YoY growth
Cash Flow	L1	Change in deposits
Cash Flow	L2	Decrease in deposits

Section	Level	Metric
Cash Flow	L2	Increase in deposits
Cash Flow	L1	Issuance/retirement of stock (net)
Cash Flow	L2	Repurchase of common & preferred stock
Cash Flow	L2	Sale of common & preferred stock
Cash Flow	L3	Other Proceeds from Sale/Issuance of Sto...
Cash Flow	L3	Proceeds from stock options
Cash Flow	L1	Total cash dividends paid
Cash Flow	L1	Issuance/retirement of debt (net)
Cash Flow	L2	Issuance/retirement of short term debt
Cash Flow	L2	Issuance/retirement of long term debt
Cash Flow	L3	Issuance of long term debt
Cash Flow	L3	Reduction of long term debt
Cash Flow	L2	Issuance/retirement of other debt
Cash Flow	L1	Other financing cash flow items (total)
Cash Flow	L2	Financing activities (other uses)
Cash Flow	L2	Financing activities (other sources)
Cash Flow	L1	Exchange rate effect
Cash Flow	L1	Miscellaneous funds
Cash Flow	L1	Net change in cash
Cash Flow	L1	Free cash flow // YoY growth
Cash Flow	L1	Interest paid
Cash Flow	L1	Taxes paid

SCHEMA_02_STATISTICS (Quarterly)

Group	Metric
Valuation ratios	Price to earnings ratio
Valuation ratios	Price to sales ratio
Valuation ratios	Price to cash flow ratio
Valuation ratios	Price to book ratio
Valuation ratios	Enterprise value
Valuation ratios	Enterprise value to EBITDA ratio
Profitability ratios	Return on assets %
Profitability ratios	Return on equity %
Profitability ratios	Return on invested capital %
Profitability ratios	Gross margin %
Profitability ratios	Operating margin %
Profitability ratios	EBITDA margin %
Profitability ratios	Net margin %
Liquidity ratios	Asset turnover
Solvency ratios	Debt to assets ratio
Solvency ratios	Debt to equity ratio
Solvency ratios	Long term debt to total assets ratio
Solvency ratios	Long term debt to total equity ratio
Per share metrics	Revenue per share
Per share metrics	Operating cash flow per share
Per share metrics	Free cash flow per share
Per share metrics	EBIT per share
Per share metrics	EBITDA per share
Per share metrics	Book value per share

Group	Metric
Per share metrics	Total debt per share
Per share metrics	Cash per share
Per share metrics	Net current asset value per share
Per share metrics	Tangible book value per share
Per share metrics	Working capital per share
Per share metrics	CapEx per share

SCHEMA_03_DIVIDENDS (Annual)

Metric
Dividends per share (FY)
Dividend yield (FY) %
Payout ratio (FY) %

SCHEMA_04_EARNINGS (Quarterly)

Block	Fields
EPS	Reported, Estimate, Surprise
Revenue	Reported, Estimate, Surprise

SCHEMA_05_REVENUE (Annual)

Block	Fields	Examples
By Source	Source, Amount (Millions)	VivoCity, Mapletree Business City, Festival Walk, China Properties, Other Singapore Properties, Japan Properties, MTower, Mapletree Anson, Merrill Lynch HarbourFront, Retail, Office
By Country	Source, Amount (Millions)	Singapore, Hong Kong, China, Japan, I

Appendix C: Tickers Chosen

Healthy REITs

Name	Ticker	Sector(s)	Notes
CapitaLand Integrated Commercial Trust	C38U	Retail + Office	Only 1Q and 3Q
Suntec REIT	T82U	Retail + Office	-
Mapletree Pan Asia Commercial Trust	N21U	Retail + Office	-
Fraser's Centrepoint Trust	J69U	Suburban	-
CapitaLand Ascendas REIT	A17U	Industrial	-

Name	Ticker	Sector(s)	Notes
Keppel DC REIT	AJBU	Data Center	-
Mapletree Industrial Trust	ME8U	Data Center	-
Mapletree Logistics Trust	M44U	Logistic	-
Fraser's Logistics & Commercial Trust	BUOU	Logistic	-

Distressed REITs

Name	Ticker	Sector(s)	Notes
Lippo Malls Indonesia Retail Trust	D5IU	Retail	ICR fell to 1.36x (2024) then 1.29x (early 2025); linked to higher interest + IDR weakness; MAS restriction triggered below 1.5x; major gap-down and 3 crash phases (2018/2020/2023)
Prime US REIT	OXMO	Office	ICR around 1.3x-1.7x; MAS recovery-plan disclosure trigger discussed; massive gap-down noted
Manulife US REIT	BTOU	Office	ICR toward lender limit (~2.0x), then near 1.5x; gearing exceeded 50% due to asset value decline; massive gap-down
EC World KORE / CMOU KORE US REIT	BWCL / CMOU	Logistics	Office Likely severe ratio deterioration; metric/context needs verification
Sabana	M1GL	Industrial	Does not look distressed; possibly use as healthy/control only after validation

Appendix D: AI Tools

Claude Sonnet 4.6 was used for AI-assisted drafting workflow.

Appendix E: Acknowledgements

N/A